

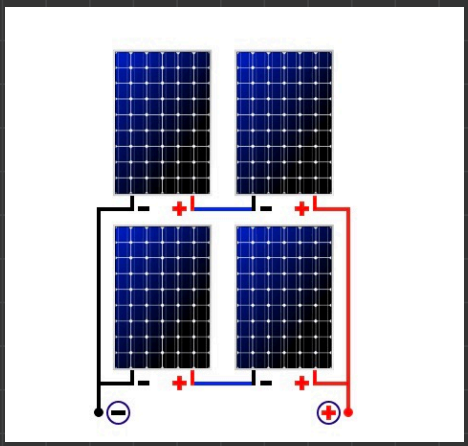
Project Details

- Combine 4 Solar Cells
- 12 watts total
- 12V
- 1 A

1st Idea:

Series - Parallel Combo:

- 2 Series Pairs in Parallel



$V_{tot} = 12V$ ✓

$i_{tot} = 1\ A$ ✓

$P_{tot} = (12V)(1A) = 12W$ ✓

Each Solar Cell:

- 6V
- 500 mA
- 3 W

Within Configuration:

$P_{each} = (6V)(0.5A) = 3W$

$V_{series\,1} = V_1 + V_2 = 6V + 6V = 12V$

$i_{series\,1} = 0.5\ A$

$P_{series\,1} = (12V)(0.5A) = 6\ W$

$V_{series\,2} = 12V$

$i_{series\,2} = 0.5\ A$

$P_{series\,2} = (12V)(0.5A) = 6W$

2nd Idea:

All Parallel Combo:

- 4 solar cells in Parallel (12 V, 250 mA e/a)

* NOT recommened since doesn't give flexability in designing converter, doesn't allow us to change voltage (just manipulating current)

Concerns:

1) Actual Operating voltage vs Advertised voltage

- Panel advertised as 6 V won't necessarily output 6 V
- Look for $V_{max\,power}$ and $V_{open\,circuit}$
- Converter should be able to handle range of volts. the Panels in the series-Parallel config. can produce

2) Actual Power Output

- The advertised Power Output is usually under specific tests/conditions
- light intensity, Panel angle, Shade, temperature can affect Power
- Don't design converter assuming it'll always recieve 12 V.

3) Current variation

- Advertised Current not always guaranteed
- converter should work if Panels Produce much less than 1 A